

Week 5 Assignment

Repository Link:

<https://github.com/Hughes-Tyler/AnalyticsCapstone>

Report Link:

<https://github.com/Hughes-Tyler/AnalyticsCapstone/blob/main/notebooks/main.ipynb>

This week we prepared the cleaned startup dataset for modeling. We engineered two new features, decided how to handle the heavy class imbalance, reduced the numeric columns with principal component analysis, and assembled the final dataset.

We created two features that combine existing columns. The first, log average funding per round, divides total funding by the number of funding rounds. Two companies can raise the same total, but one may have done it in a single large round and the other in several small ones. That difference says something about investor confidence and company trajectory that neither column shows alone. The second, log rounds per year, divides the number of rounds by the length of the fundraising window in years, capturing how quickly a company was raising money. A company that raised four rounds in one year is on a very different path from one that raised four rounds over six years. Both were computed in log space to match our other numeric features. Funding per round is imputed for about 18% of companies whose funding total was missing, but the `funding_missing` flag stays in the dataset so the model can account for it. The pace feature is weaker, because about 59% of companies raised only one round and therefore all share a value of 0.

The target is heavily imbalanced: 81.7% of companies are operating, 8.3% acquired, 8.0% closed, and only 2.0% IPO. Before deciding how to handle this, we created a stratified 60/20/20 train, validation, and test split and saved only the row indices, so the underlying data was not changed. The training set has 29,782 companies, including 592 IPOs. We decided not to use SMOTE. Of our 53 features, 46 are 0/1 columns, and when we generated 500 synthetic IPO rows the way SMOTE does, 89.4% contained at least one invalid value, such as 0.6 for a country flag. SMOTE also assumes that a company between two IPO companies would itself go public, which is hard to support because our labels are a snapshot in time. Instead we will use balanced class weights, computed from the training labels only, which give IPO a weight of

12.58 compared to 0.31 for operating. Every row remains a real company, and we will evaluate models using macro F1 and per-class recall rather than accuracy.

For dimensionality reduction, we ran PCA only on the 7 continuous numeric columns: funding rounds, founded year, the three log-transformed funding and timing columns, and the two new features. We excluded the 46 binary country, category, and flag columns because PCA on sparse 0/1 data is hard to interpret and would blur the geographic and industry signals. The scaler and PCA were fit on the training rows only and then applied to the whole dataset, which prevents information from the validation and test sets from leaking into the components. Five components were enough to capture 95% of the variance in the numeric columns.

The final dataset replaces the 7 numeric columns with the five principal components and keeps the 46 binary columns and the status target unchanged. It contains 49,638 rows and 52 columns, with no missing values. No rows were dropped and the index was never reset, so the train, validation, and test splits still line up with the same companies. We verified this with checks on the row counts, index, target, and missing values. The main tradeoff is interpretability: the principal components are harder to explain than the original numeric columns, so we may compare against a model that keeps the raw numeric features.